# Dual Airfoil Testing Platform


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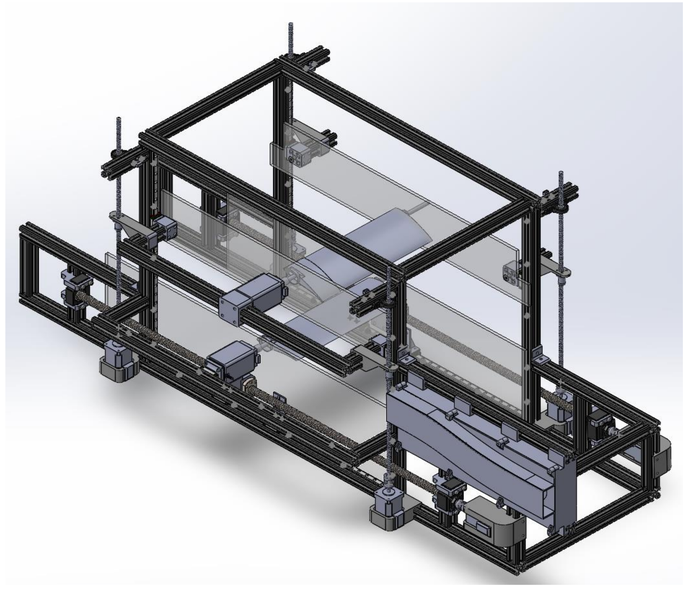
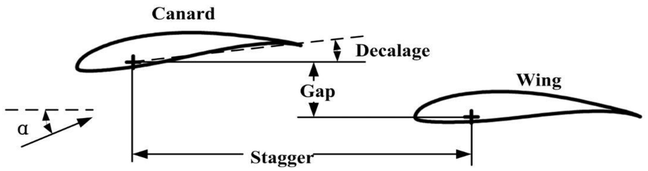
## Objective

The objective of this project is to improve the previous group's design. This was primarily done by decreasing the lose wires by implementing a Bluetooth connection and adding encoders to the motors to continuously read the position of the motors. This prevents motors from missing their marks or becoming too out of sync.



## Why Dual Airfoils?

Dual airfoils can both increase lift and reduce the likelihood of stall. The flow characteristics caused by the airfoils is dependent on the airfoil positions. This creates the possibility for future aircraft to change airfoil positions during flight for optimal performance.




## Decision Matrices

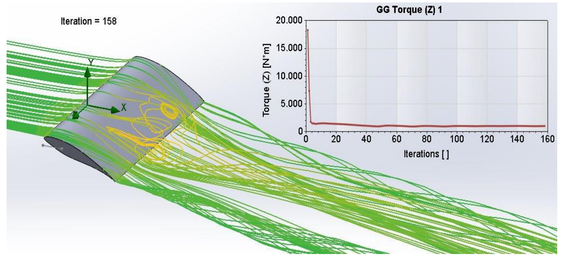
Used to determine what the measurement device should be for the x and y –axis motors along with the AoA motors. Another matrix was created to choose how to reduce the wiring.

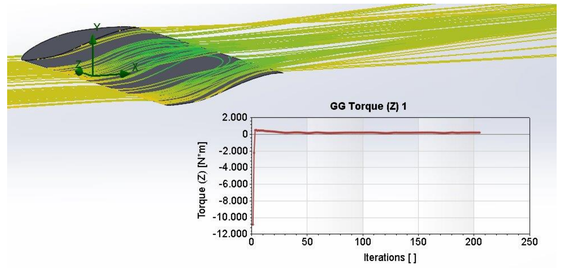
| Layout | Wiring | Power Sources | Wireless Connections | Ease of Implementation | Cost | Total |
|---|---|---|---|---|---|---|
| 1 board per device | 5/5 | 1/5 | 1/5 | 2/5 | 1/5 | 10/25 |
| Single main board | 1/5 | 5/5 | 5/5 | 4/5 | 5/5 | 20/25 |

| Design | Accuracy | Precision | Ease of Implementation | Cost | Total |
|---|---|---|---|---|---|
| TOF Sensor | 1/5 | 1/5 | 4/5 | 5/5 | 11/20 |
| Absolute Encoder | 5/5 | 5/5 | 3/5 | 1/5 | 14/20 |
| Incremental Encoder | 5/5 | 5/5 | 4/5 | 3/5 | 17/20 |

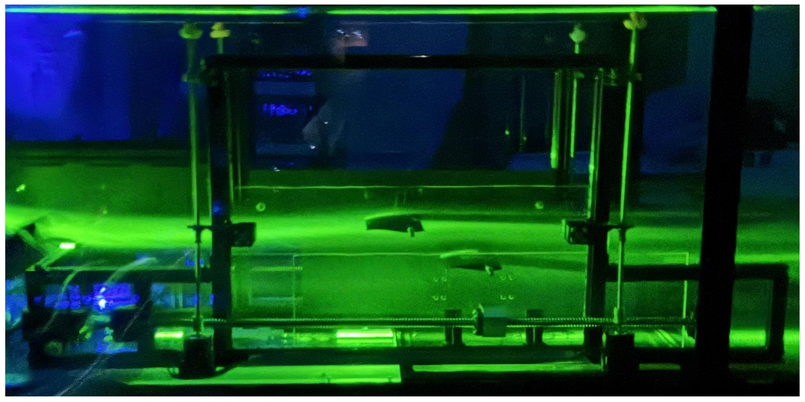
| Design | Line of Sight | Ease of Implementation | Adaptability | Total |
|---|---|---|---|---|
| Encoder opposite to motor | 1/5 | 2/5 | 4/5 | 7/15 |
| Encoder (through-bore) stacked on motor | 5/5 | 4/5 | 5/5 | 14/15 |
| Motor-encoder Combo | 5/5 | 4/5 | 1/5 | 10/15 |

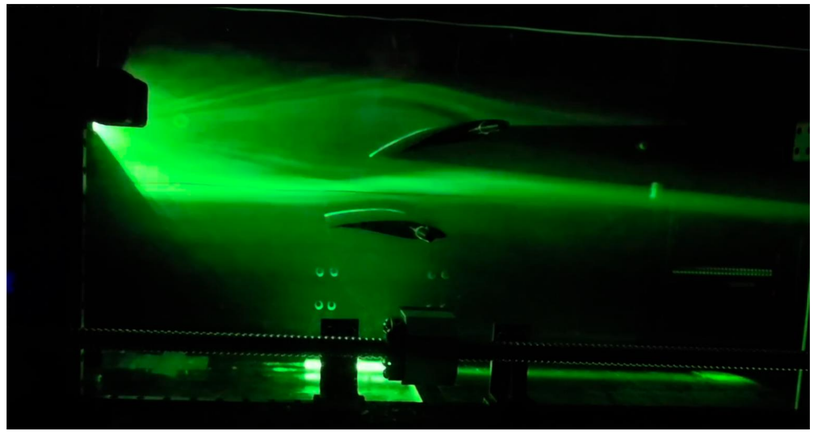
## CFD Analysis



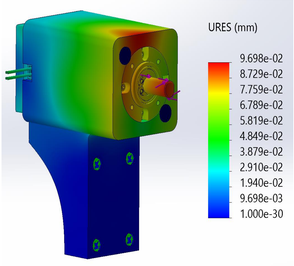



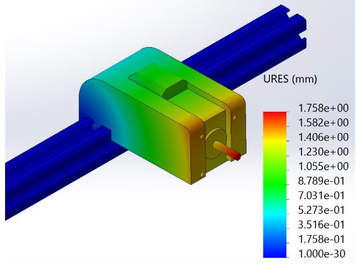

## Flow Visualization

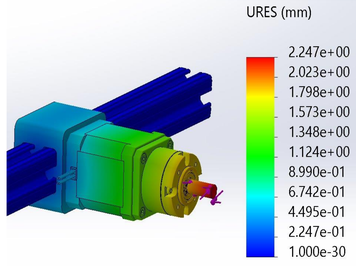




## FEA




FEA was performed to confirm that our new motor mounts would not break or deform under the loading. A CFD analysis was also performed to verify the amount of torque the motors needed to withstand under wind tunnel testing.